



Tutorial S2B: Binomial Distribution

Section A (Basic Questions)

1 Given $X \sim B(10, 0.2)$, find

- (i) $P(X = 0)$,
- (ii) $P(X \leq 2)$,
- (iii) $P(X \geq 5)$,
- (iv) $P(4 \leq X \leq 8)$,
- (v) $P(2 \leq X < 6)$,
- (vi) $E(X)$ and $\text{Var}(X)$.

[**(i)** 0.107 **(ii)** 0.678 **(iii)** 0.0328 **(iv)** 0.121 **(v)** 0.618 **(vi)** 2, 1.6]

Solution:	
(i)	From the GC, $P(X = 0) \approx 0.10737 = 0.107$ (3 s.f.) Note: use the binompdf (command in GC
(ii)	From the GC, $P(X \leq 2) \approx 0.6777995 = 0.678$ (3 s.f.) Note: use the binomcdf (command in GC
(iii)	$P(X \geq 5) = 1 - P(X \leq 4) \approx 0.032793 = 0.0328$ (3 s.f.)
(iv)	$P(4 \leq X \leq 8) = P(X \leq 8) - P(X \leq 3) \approx 0.12087 = 0.121$ (3 s.f.)
(v)	$P(2 \leq X < 6)$ $= P(2 \leq X \leq 5)$ $= P(X \leq 5) - P(X \leq 1)$ ≈ 0.61782 $= 0.618$ (3 s.f.)
(vi)	$E(X) = np = (10)(0.2) = 2$ $\text{Var}(X) = np(1 - p) = (10)(0.2)(1 - 0.2) = 1.6$

Formula for mean and variance
can be found in MF27.

- 2 A die is biased such that the probability of getting a six is 0.18. If the die is thrown ten times, find
- (i) the probability that the number of sixes that turn up is odd,

- (ii) the expectation and variance of the number of sixes that turn up.

[(i) 0.494 (ii) 1.8, 1.476]

Solution:

- (i) Let X be the **number** of sixes obtained, out of 10 throws.

Then $X \sim B(10, 0.18)$

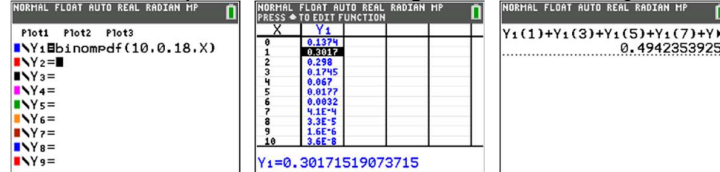
$P(X \text{ is odd})$

$$= P(X=1) + P(X=3) + P(X=5) + P(X=7) + P(X=9)$$

$$\approx 0.49424 = 0.494 \text{ (3 s.f.)}$$

Note that it is compulsory to define the random variable and write down its distribution.

Note: You may wish to table the probabilities using the GC as shown.



- (ii) $E(X) = np$ $\text{Var}(X) = np(1-p)$
- $$= 10(0.18) = 10(0.18)(1-0.18)$$
- $$= 1.8 = 1.476$$

Expectation and variance of number of sixes that turn up is 1.8 and 1.476 respectively.

- 3 Two fair dice are thrown. Find the probability that the total score is two-figured.

If five people were each to throw two dice, find the probability that exactly three of them will get a two-figured total score.

$[\frac{1}{6}, 0.0322]$

Solution:

For total score to be two-figured, the possible outcomes for the two dice are: (4, 6), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6).

$$\text{Probability that total score is two-figured} = \frac{6}{36} = \frac{1}{6}.$$

Let Y be the **number** of people, out of the 5, who get a two-figured total score.

$$Y \sim B\left(5, \frac{1}{6}\right)$$

$$P(Y=3) \approx 0.032150 = 0.032 \text{ (3 s.f.)}$$

Pay attention to how a random variable which follows Binomial Distribution is defined.

- 4 ‘Droppers’ are small sweets that are made in a variety of colours. Droppers are sold in packets and the colours of the sweets in a packet are independent of each other. On average, 25% of Droppers are red. A small packet of Droppers contains 10 sweets.

- (i) Find the probability that there are at least 4 red sweets in a small packet.
- (ii) Yip buys 15 small packets of Droppers. Find the probability that no more than 3 of these packets contain at least 4 red sweets.

[(i) 0.224 (ii) 0.557]

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Solution:

- (i) Let X be the **number** of red sweets in a small packet of 10 sweets.
Then $X \sim B(10, 0.25)$
 $P(X \geq 4) = 1 - P(X \leq 3)$
 $\approx 1 - 0.775875$
 $= 0.224125$
 $= 0.224$ (3 s.f.)
- (ii) Let W be the **number** of small packets, out of 15, containing at least 4 red sweets.
 $W \sim B(15, P(X \geq 4))$
That is, $W \sim B(15, 0.224125)$
 $P(W \leq 3) \approx 0.557051 = 0.557$ (3 s.f.)

Use appropriate accuracy (at least 5 s.f.)
for the value of p

THE END